

① Image Compression

Representing same or almost same image with less no. of bits
without significant degradation

→ To reduce storage requirement

→ To reduce no. of bits

Metric Relative data redundancy, $R = 1 - \frac{1}{C}$

Original size $\rightarrow B$ bits

Compressed size $\rightarrow k$ bits

Compression Ratio $C: B/k (> 1)$

HIGHER IS BETTER

% REDUCTION: $\frac{B-k}{B} \times 100\%$

HIGHER IS BETTER

bpp (bits per pixel) $\frac{k}{M \times N}$ image dimension $(M \times N)$
(LOWER IS BETTER)

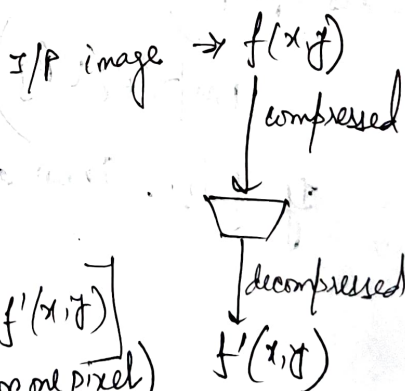
$$\begin{aligned} R_{Q.B} &= 1024 \times 512 \times 3 \text{ bytes} \\ &= 2^{10} \times 2^8 \times 2^9 \\ &\approx 2^{28} \text{ bits} = 2^{20} \text{ bytes} \\ &\text{transmit in } 2 \text{ sec} \\ &= 2^{10} \text{ bytes/sec.} \end{aligned}$$

IMAGE $\xrightarrow{\text{COMPRESS}}$ COMPRESSED IMAGE
 $\xleftarrow{\text{DECOMPRESS}}$

COMPRESSION

LOSS-LESS

LOSSY



(for all pixels)

$$\left[f(x,y) = f'(x,y) \right] \quad \left[f(x,y) \neq f'(x,y) \right] \quad \text{(at least for one pixel)}$$

$$f(x,y) \xrightarrow{\text{COMP}} \boxed{} \xrightarrow{\text{DECOMP}} f'(x,y)$$

Quality error: $|f(x,y) - f'(x,y)|$

$$\text{Total Error} = \sum_x \sum_y |f(x,y) - f'(x,y)|$$

②

The term data compression refers to the process of reducing the amount of data required to represent a given quantity of information.

Data and information are not the same. Data are the means by which information is conveyed. Representations that contain irrelevant or repeated information are said to contain redundant data.

Two dimensional intensity arrays suffer from 3 principal types of data redundancies —

① Coding Redundancy +

A code is a system of symbols used to represent a body of information or set of events. Each piece of information or event is assigned a sequence of code symbols, called a codeword. The no. of symbols in each code word is its length.

The 8-bit codes that are used to represent the intensities in most arrays contain more bits than are needed to represent the intensities.

② Spatial and Temporal (Interpixel) Redundancy +

Because the pixels of most 2-D intensity arrays are correlated spatially i.e. each pixel is similar to or dependent upon neighbouring pixels, the information is unnecessarily replicated in the representations of the correlated pixels.

③ Psycho-Visual Redundancy +

Most 2D intensity arrays contain information that is ignored by the human visual system and/or extraneous to the intended use of the image. It is redundant in the sense that it is not used.

9) let $r_k \rightarrow$ discrete random variable in interval $[0, L-1]$ used to represent the intensities of an $M \times N$ image and that each r_k occurs with probability $P_r(r_k)$

$$P_r(r_k) = \frac{n_k}{MN} \quad [k=0, 1, \dots, L-1]$$

$n_k =$ no. of times k th intensity appears in the image.

$l(r_k) \rightarrow$ bits used to represent each value of r_k

$$L_{avg} = \sum_{k=0}^{L-1} l(r_k) P_r(r_k)$$

The total no. of bits required to represent an $M \times N$ image - MNL_{avg}

Shannon's first Theorem/noiseless coding theorem $\Rightarrow \lim_{n \rightarrow \infty} \left[\frac{L_{avg} \cdot n}{n} \right] = H$

where H is entropy, $H = - \sum_{k=0}^{L-1} P_r(r_k) \log_2 P_r(r_k)$

The removal of 'irrelevant visual' information involves a loss of real or quantitative image information. As information is lost, a means of quantifying the nature of the loss is needed. Two types of criteria can be used for such an assessment -

- (i) Objective fidelity criteria
- (ii) Subjective " "

When information loss can be expressed as a mathematical function of the input and output of a compression process, it is said to be based on an objective fidelity criterion.

$$e(x, y) = \hat{f}(x, y) - f(x, y)$$

$$\text{total error} = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\hat{f}(x, y) - f(x, y)]$$

$$R_{rms} = \left[\frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\hat{f}(x, y) - f(x, y)]^2 \right]^{1/2}$$

$$SNR_{rms} = \frac{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y)^2}{\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\hat{f}(x, y) - f(x, y)]^2}$$

$\hat{f}(x, y) \rightarrow$ approximation of $f(x, y)$ that results from compressing and subsequently decompressing the input

Root mean square error (rmse error)

$$\sqrt{\frac{1}{MN} \sum_x \sum_y (f(x,y) - f'(x,y))^2}$$

Signal to noise ratio (SNR)

$$\frac{\sum_x \sum_y f'(x,y)^2}{\sum_x \sum_y (f(x,y) - f'(x,y))^2}$$

Peak signal to noise ratio (PSNR)

$$\frac{MN \cdot 255^2}{\sum_x \sum_y (f(x,y) - f'(x,y))^2}$$

COMPRESSION



Exploits redundancy

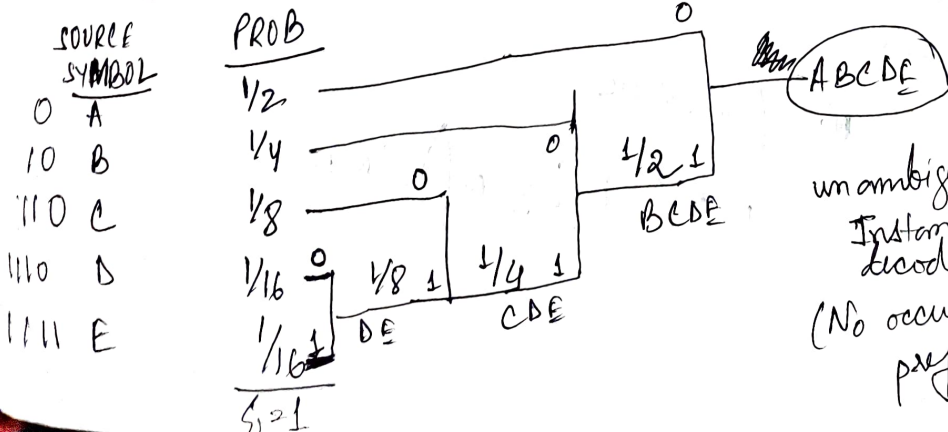
- Coding Redundancy
- Interpixel Redundancy
- Psychovisual Redundancy

Coding Redundancy

Coding → Fixed length encoding
 → Variable length encoding
 ↓
 Symbolic separator (overhead)

Prefix Code
 ↓
 No smaller codeword is a prefix
 A longer codeword

Huffman Coding → To remove coding redundancy
 More frequent symbolic → smaller length, less likely → larger codeword



unambiguous/
 instantaneous
 decoding
 (No occurrence of
 prefix)

(5)

$$\bar{L} = \sum_i P_i l_i$$

$$= 1 \cdot \frac{1}{2} + \frac{1}{4} \times 2 + \frac{1}{8} \times 3 + \frac{1}{16} \times 4 + \frac{1}{16} \times 4$$

$$= \frac{1}{2} + \frac{1}{2} + \frac{3}{8} + \frac{1}{2}$$

$$= \frac{4+4+3+4}{8} = \frac{15}{8}$$

Shannon's Theorem

$$\bar{L} \geq H(X)$$

If the symbols are not equiprobable $\rightarrow$ varying prob.
 $\rightarrow$ unequal distribution

Interpixel Redundancy:-

A Pixel's intensity value can be well approximated from its neighbouring values.

high correlation exists with neighbour

Individual pixel intensity $\rightarrow$ exists redundancy in information



Get rid of this redundancy.
 No need to store intensity for each pixel.

Binary Image:-

0, 1

Run length encoding

Run $\rightarrow$ A sequence of same value consecutive pixels with same value $\rightarrow$ Run

for each row

(0, n1) (1, n2)

(value, run length)
 1 bit $\lceil \log_2 N \rceil$

No. of Runs $\rightarrow K$

$$K * (1 + \lceil \log_2 N \rceil) < MN$$

⑥

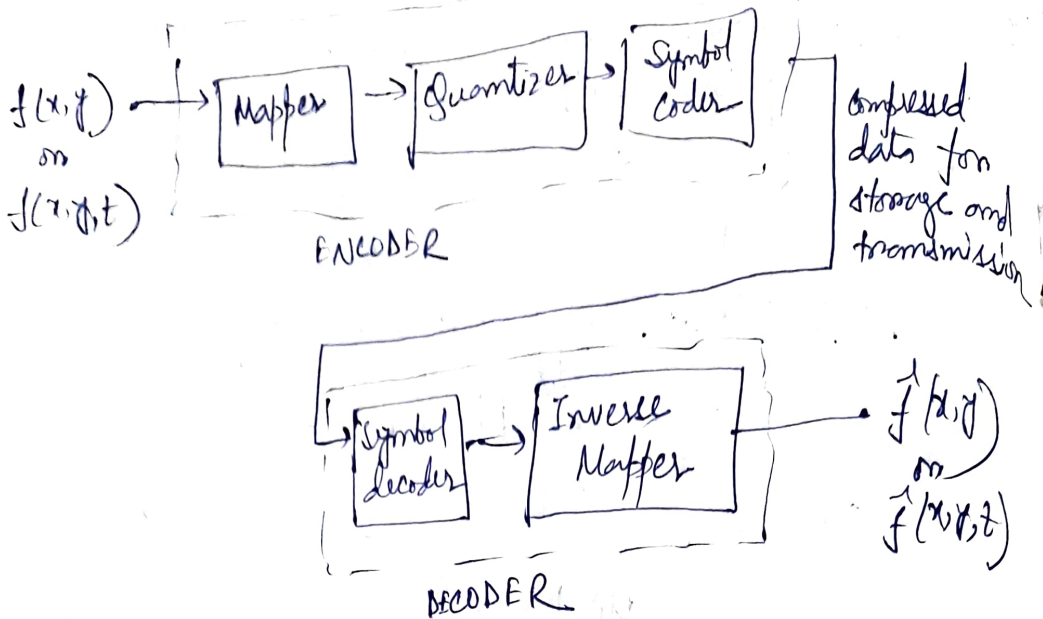


Image Compression Models

In general, $\hat{f}(x, \dots)$ may or may not be an exact replica of $f(x, \dots)$. If it is, then the compression system is called error-free, lossless, or information preserving.

If not, the reconstructed output image is distorted and the compression system is referred to as lossy.

The encoding or Compression process:-

The encoder is designed to remove the redundancies, through a series of 3 independent operations. In the first stage of the encoding process, a mapper transforms $f(x, \dots)$ into a format designed to reduce spatial and temporal redundancy. This operation is generally reversible. Run-length encoding is an example of a mapping that normally yields compression in the first step of the encoding process.

The quantizer reduces the accuracy of the mapper's output in accordance with a pre-established fidelity criterion. The goal is to keep irrelevant information out of the compressed representation. This operation is irreversible. It must be omitted when error-free compression is desired.

⑦ Symbol Codes generate a fixed-length or variable length code to represent the quantizer output, and maps the outputs in accordance with the code. In many cases, a variable length code is used, the shortest codewords are assigned to the most frequently occurring quantizer output values, thus minimizing coding redundancy. This operation is reversible.

The decoding or Decompression Process

The decoder contains only two components — a symbol decoder and an inverse mapper. They perform, in reverse order, the inverse operations of the encoder's symbol encoder and mapper. Because quantization results in irreversible information loss, an inverse quantizer block is not included in the general decoder model.

Huffman Coding

One of the most popular techniques for removing coding redundancy is due to Huffman. When coding the symbols of an information source, individually, Huffman coding yields the smallest possible no. of code symbols per source symbol.

In terms of Shannon's first theorem, the resulting code is optimal for a fixed value of n , subject to the constraint that the source symbols be coded one at a time.

First Step, Create a series of source reductions by ordering the probabilities of the symbols under consideration, then combining the lowest probability symbols into a single symbol that replaces them in the next source reduction. This process is then repeated until a reduced source with two symbols is reached.

Second Step, Code each reduced source, starting with the smallest source and working back to the original source. The minimal length binary code for a two symbol source, of course are the symbols 0 and 1.

⑧

Huffman's procedure creates the optimal code for a set of symbols and probabilities subject to the constraint that the symbols be coded one at a time. After the code has been created, coding and/or error-free decoding is accomplished in a simple look-up table manner. The code itself is an instantaneous uniquely decodable block code. It is called a block code because each source symbol is mapped into a fixed sequence of code symbols. It is instantaneous because each code word in a string of code symbols can be decoded without referencing succeeding symbols. It is uniquely decodable as any string of code symbols can be decoded in only one way.

⑨ Run-length Coding

Images with repeating intensities along their rows (or columns) can often be compressed by representing runs of identical ~~repeated~~ intensities as run-length pairs, where each run-length pair specifies the start of a new intensity and the number of consecutive pixels that have that intensity.

Compression is achieved by eliminating a simple form of spatial redundancy - groups of identical intensities. When there are few (or no) runs of identical pixels, run-length encoding results in data expansion.

It is particularly effective when compressing binary images, because there are only two possible intensities, adjacent pixels are more likely to be identical. The basic idea is to code each contiguous group of 0's and 1's encountered in a left-to-right scan of a row by its length and to establish a convention for determining the value of the run.

① The most common conventions are (1) to specify the value of the first run of each row, or (2) to assume that each row begins with a white run, whose run length may be in fact be zero,

$$\text{Entropy, } H_L = \frac{H_0 + H_1}{L_0 + L_1}$$

$L_i \rightarrow$ avg values of black and white run lengths.

second byte value

0

1

2

3-255

condition

End of line

End of image

Move to a new position

specify pixels individually

Psycho-Visual Redundancy:-

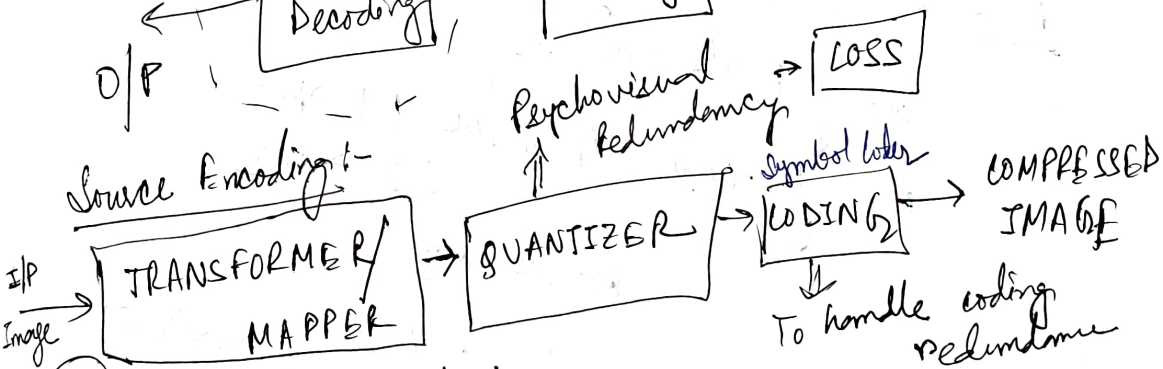
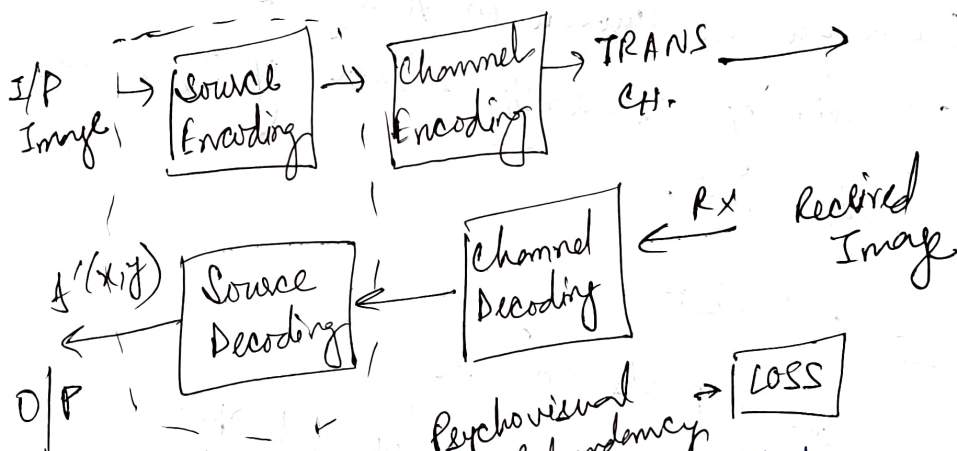
Human eye is not sensitive to every pixel information, it does not perform quantitative analysis of each pixel. Why to store all?

Eye is sensitive to only distinctive features like line, texture etc.

Mentally combines into Recognizable groupings

Such groupings are finally help to interpret the image with prior knowledge.

→ losing certain information → LOSSY
Quantization
256 → 16 levels.
IRREVERSIBLE



Source Encoding:-

I/P Image

(P/E)

Interpixel Redundancy

Normally, Reversible may or may not compress
GRAYSCALE IMAGE → LOSS-LESS (NO QUANTIZATION)

for r , 2nd row to last row

$$g(r, c) = g(r-1, c) - g(r, c) \text{ for all columns}$$

(11)

lossy

Block truncation coding (BTC)

→ Image is divided into $m \times m$ non overlapped blocks.

→ Pixels in each block are represented by either of two labels L_1 & L_2 (say L_1, L_2)

 L_1 & L_2

threshold
Pixels with value $\leq th \rightarrow L_1$
else L_2

L_1 & L_2 are chosen in a way such that avg intensity & intensity variance of the block remain same.

For a block

No. of pixels $\Rightarrow m^2 = M$

No. of pixels with value $L_1 \Rightarrow M_1$
 $L_2 \Rightarrow M_2$

$$M = M_1 + M_2$$

$$\frac{1}{M} (M_1 L_1 + M_2 L_2) = \frac{1}{M} \sum_x \sum_y f_i(x, y) = \bar{f}_i \rightarrow \text{eq 1}$$

$$\bar{f}_i^2 = \frac{1}{M} \sum_x \sum_y f_i^2(x, y)$$

$$M_1 (L_1 - \bar{f}_i)^2 + M_2 (L_2 - \bar{f}_i)^2 = \sum_x \sum_y (f_i(x, y) - \bar{f}_i)^2$$

$$\Rightarrow M_1 L_1^2 + M_2 L_2^2 - 2M_1 L_1 \bar{f}_i - 2M_2 L_2 \bar{f}_i + M \bar{f}_i^2 = \sum_x \sum_y f_i^2(x, y) - 2M \bar{f}_i \bar{f}_i + M \bar{f}_i^2$$

$$= M \bar{f}_i^2 + M \bar{f}_i^2 - 2M \bar{f}_i \bar{f}_i$$

$$= M \bar{g}_i^2 + M \bar{g}_i^2 - 2M \bar{g}_i \bar{g}_i$$

$$\rightarrow M_1 L_1^2 + M_2 L_2^2 = M \bar{g}_i^2 \Rightarrow \text{eq (2)}$$

$$\hookrightarrow \frac{1}{M} (M_1 l_1^2 + M_2 l_2^2) = \overline{I_i^2}$$

$$L_1 = \bar{g}_i - \gamma \sqrt{M_2/M_1}$$

$$r = \sqrt{\bar{g}_i^2 - \bar{g}_i^2}$$

$$L_2 = \overline{g_i} + \sqrt{M_2/M_1}$$

ith Block

$\frac{k^2}{m^2} (m)$ intensity values
↓ ORDER THEM IN ASC—

$$g_i(0)$$
$$g_i(t)$$
 γ
$$g(M-1)$$
$$\text{threshold} = g_i(k)$$
$$\therefore f_j(x) - f_j(x-1)$$
$$y_{g_i(j)} - g_i(j-1)$$

for all $j \neq k$

else

Fi

for each block $\rightarrow$ $\begin{matrix} L1 & L2 \\ 1 & 2 \end{matrix}$

for each pixel $\rightarrow 0/1$

each pixel $\rightarrow 0/1$
in the block After compression $\rightarrow (2ab + M)$ bits / block

Original $\rightarrow$ 16 bits/Block

for intensity value $\Rightarrow$ 6 bits